CS 520 Final Project Report

Decision-Theoretic Navigation Under Uncertainty in Dynamic Grid Environments

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Abstract

We study decision-theoretic navigation under uncertainty by pitting four families of agents — static A*, replanning A*, risk-aware A*, and tabular Q-learning — against the canonical Pac-Man maze, which we treat as a dynamic grid with mobile, partially-observable hazards (ghosts). Each agent is run for 50 Monte-Carlo episodes on a shared seed pool, and we report mean score, level-clear rate, time-to-failure, hazard exposure, and per-episode score distributions. Q-learning leads the field on score (3,249–3,757 across the three death-policy variants), narrowly above replanning A* and risk-aware A* (both ≈3,150 across $\lambda \in \{0.5, 1, 2, 5\}$). Static A* without replanning collapses (1,479 mean score, 24-step time-to-failure), confirming that probabilistic replanning is essential under mobile hazards. The risk-aware extension does not move the needle on this specific maze because hard ghost-obstacle culling already keeps the planned path far from any ghost where the additive `λ·E[risk]` term can fire. The Q-learning agents earn *more* score than the planners while spending 1.4–5× less time within ghost-radius 2, suggesting they have learned a value function that internalises ghost-avoidance. Heavier death penalty (model 1 → 3) yields a stronger learned policy on every metric we measured. We close with a quantitative discussion of when the decision-theoretic term matters and when it does not.

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1. Introduction

Real-world AI agents — autonomous vehicles, mobile robots, simulated game characters — rarely operate in static environments. Hazards move, evolve, or spread, and the agent must plan paths while reasoning about *where the hazards will be* by the time the agent arrives. Classical shortest-path algorithms such as BFS or A* assume a static world and produce paths that become stale the moment the world changes.

This project asks a focused question: **how much does explicit reasoning about future hazard locations help an agent navigating a dynamic grid, and how does it compare to (a) reactive replanning, (b) hand-tuned heuristic behaviour, and (c) reinforcement-learning?**

We answer with four agent families, a fixed Monte-Carlo evaluation harness, and ∼50 episodes per agent on a paired seed pool, producing the analytics in Section [§8].

2. Background and Prior Work

The proposal cites the foundational A* paper of Hart, Nilsson, and Raphael (1968) [1], whose admissible-heuristic framework underpins three of our four agents. Beyond classical search, we draw on:

- **Decision-theoretic planning.** Adding an expected-cost term to A*'s evaluation function turns a deterministic shortest-path problem into a utility-maximisation problem. We use the canonical form `f(n) = g(n) + h(n) + λ · E[risk(n)]`.
- **Tabular reinforcement learning** (Watkins, 1989; Sutton & Barto, 2018). Q-learning is appropriate when the dynamics model is unknown or prohibitively expensive to query online, and it gives us a learned baseline against which to measure the value of explicit planning.
- **Pac-Man as an AI benchmark.** Pac-Man has been used as a teaching and benchmarking environment in classic AI courses (UC Berkeley CS188) and in reinforcement-learning literature precisely because it combines mobile adversarial agents, sparse rewards, and a fixed maze topology — which is exactly the setting our proposal targets.

3. From Fire-Spread Grid to Pac-Man — Project Scope

The proposal originally framed the environment as a "spacecraft grid" with hazards that *spread stochastically* and stay hazardous once burning. During implementation we pivoted the surface form of the environment to the canonical 28×36 Pac-Man maze for three pragmatic reasons:

1. **Mobile hazards strictly generalise spreading hazards** for the agent's decision problem. Both share the same abstract structure: at any given timestep some set of cells is unsafe; at the next timestep the unsafe set is a stochastic function of the current one. The risk-aware planning math is identical.
2. **Pac-Man already ships with a faithful, deterministic-given-seed Python simulator** (`HighScorePacmanEnv`), which lets us close the experimental loop in days rather than weeks.
3. **Recognisability.** Reviewers have strong priors about what good Pac-Man play looks like, which makes plots much easier to read.

The mapping is direct:

Proposal concept	Pac-Man instance
Open / blocked cells	Walkable tiles / wall tiles
Goal cell	"Eat all pellets" (clear the level)
Hazardous cells	Active non-scared ghost cells (and their 1-step neighbourhood)
Stochastic spread of hazard	Stochastic ghost movement (uniform-on-legal-moves model used by the risk-aware agent)
Hazard sensitivity λ	Same λ in `f(n) = g(n) + h(n) + λ · E[risk(n)]`

Every proposal point — three search agents, the decision-theoretic formula, Monte Carlo evaluation across success rate, path length, and time-to-failure — is realised against this environment. We additionally added a reinforcement-learning track and two sanity baselines (random + a hand-coded heuristic) to give the search agents a richer comparison.

4. Environment

The Pac-Man simulator (`RL/rl_q_learning.py:HighScorePacmanEnv`) loads the canonical 28×36 layout, parses pellets, energizers, and walls, and pre-computes BFS shortest-path distances between every pair of walkable tiles. Each episode begins with 244 pellets, 4 energizers, 4 ghosts, and 3 Pac-Man lives. The agent submits one of `{UP, LEFT, DOWN, RIGHT}` per timestep; the env applies the move, awards points (10 / 50 / fruit-bonus / ghost-combo), advances ghosts, checks collisions, and emits a `StepResult` containing `reward`, `done`, `score_delta`, and `life_lost`.

Ghosts use a deterministic-given-seed greedy chase policy (manhattan distance to Pac-Man, modulo "no about-face"), while scared ghosts greedily *retreat*. This is partially-observable from the agent's point of view: the agent sees ghost positions and directions but does not know the env's internal RNG, so future ghost moves are stochastic from the agent's perspective.

Episodes terminate when (a) the agent loses its third life, (b) all pellets and energizers are eaten (level clear), or (c) the 900-step cap is reached.

5. Agents

5.1 Random baseline (`random`)

Picks uniformly among legal actions. Acts as a sanity floor.

5.2 Hand-coded heuristic (`heuristic`)

`RL/traditional_baseline.py`. Scores each candidate action with a weighted sum of nearest-ghost distance, ghost pressure, nearest-pellet BFS distance, nearest-energizer BFS distance, open-tile count, corridor depth, and a catastrophic penalty for stepping onto a ghost. This is the "classical controller" reference used by the rest of the project.

5.3 Static A* (`astar_static`) — proposal §5.1

Plans once at episode start and reuses the plan until it is invalidated. Targets the nearest pellet/energizer via BFS, then runs A* with a Manhattan heuristic. Replans only when the next step is blocked or the target tile has been consumed. **Hazards are ignored during planning**, faithfully reproducing the proposal's "Baseline Search Agent".

5.4 Replanning A* (`astar_replan`) — proposal §5.2

Replans **every step**. Treats the current ghost cell *and* its predicted 1-step neighbourhood as hard obstacles, then runs A* to the nearest reachable pellet/energizer with a unit-cost transition function. Falls back to a Manhattan-greedy step when no path exists (e.g., in fully encircled states).

5.5 Risk-Aware A* (`astar_risk`) — proposal §5.3 + §6 (the central contribution)

Same skeleton as Replanning A*, but the per-edge cost becomes:

```
cost(current, neighbor, arrival_step) = 1 + λ · E[risk(neighbor, arrival_step)]
```

where `E[risk(n, t)]` is the probability that *any* active non-scared ghost occupies tile `n` at step `t`, summed across ghosts:

```
E[risk(n, t)] = Σ_g  P(ghost_g at n after t timesteps | ghost_g at p_g now)
```

The transition kernel `P(· | ·)` is the **uniform-over-legal-moves** model: at each step every non-occupied legal neighbour is equally likely, marginal over what the real ghost AI would do. We chose this model deliberately — the proposal's §5.3 calls for *probability of becoming hazardous*, and the uniform model is exactly the maximum-entropy assumption when the agent ignores ghost identity. The implementation memoises the per-step transition distribution in `_search.py:transition_distribution`, which keeps wall-clock to roughly 0.7 s/episode despite the 900-step cap.

The full A* objective is therefore:

```
f(n) = g(n) + h(n) + λ · E[risk(n)]
```

with `h(n) = manhattan(n, target)`, matching the proposal verbatim. λ is exposed as a CLI/notebook argument so we can sweep it (Section [§8.6]).

5.6 Tabular Q-learning (`qlearning_model{1,2,3}`)

Three Q-tables learned by `RL/rl_q_learning.py` over a 10-feature bucketed state representation (Pac-Man direction, danger / edible / pellet / energizer / fruit / scared distance buckets, open-tile count, corridor depth, phase of game). The three models differ only in their **death policy**, which determines what happens to the running score after a ghost collision:

- **Model 1**: keep the score (death-as-pause). Encourages risk-tolerant pellet-grabbing.
- **Model 2**: zero the score on death. Heavily discourages dying while still leaving the surviving lives playable.
- **Model 3**: scaled penalty (`-max(floor, ratio · peak_score)` with ratchet on subsequent deaths). Most pessimistic.

At evaluation time each model picks the action with the highest Q-value under the current bucketed state; ties broken by the action ordering `UP, LEFT, DOWN, RIGHT`.

6. Mathematical Formulation

6.1 A* admissibility

Manhattan distance is admissible on a 4-connected grid (it never over-estimates the minimum step count to reach the target). Static and Replanning A* therefore enjoy the standard A* guarantee: the first path returned is optimal under unit edge costs.

6.2 Risk-aware admissibility caveat

Adding a non-negative `λ · E[risk(n)]` term to the *cost* — not the heuristic — preserves admissibility of the heuristic but changes the optimisation target from "shortest path" to "shortest expected-risk-discounted path". The agent is no longer guaranteed to minimise step count; it is guaranteed to minimise the weighted utility, which is what we want.

6.3 Risk model and complexity

The transition distribution at step `t` is `P_t = T^t δ_{p_g}` where `T` is the row-stochastic neighbour-uniform matrix and `δ_{p_g}` is the one-hot at the ghost's current position. We compute `P_t` lazily and cache by `(p_g, t)` so each episode only ever performs `O(steps_planned · |neighbours|)` matrix work in the worst case.

6.4 Q-learning Bellman update

Standard ε-greedy tabular Q-learning:

```
Q(s, a) ← Q(s, a) + α · [r + γ · max_{a'} Q(s', a') − Q(s, a)]
```

with $\alpha=0.18$, $\gamma=0.95$, ε decayed from 0.12 by $\times 0.92$ every 20% of training episodes (12,000 episodes per model). Reward shaping adds small positive bonuses for "moved away from threat", "moved toward edible ghost", and "moved toward fruit" on top of the raw score delta, plus a 500-point level-clear bonus and a 250-point timeout penalty.

7. Experimental Setup

- **Episodes per agent**: 50 (Monte-Carlo).
- **Seed pool**: `{1000, 1001, …, 1049}` shared across *all* agents, so any agent-vs-agent comparison is a paired comparison on identical initial conditions.
- **Env death policy**: per-agent. Each Q-learning model is evaluated under the policy it was trained for (model 1, 2, or 3). Non-RL agents (random, heuristic, all A* variants) use the neutral `model 1` default — their algorithms are independent of how the env resets the score on death.
- **Step cap**: 900 steps per episode (the value used by the trainer).
- **Hazard-exposure metric**: cumulative count of timesteps in which Pac-Man is within Manhattan distance 2 of *any* active non-scared ghost. Deterministic, easy to interpret, and a defensible proxy for the "expected hazard exposure" the proposal calls out.
- **Time-to-failure**: number of steps until the *first* life is lost (capped at episode length if the agent never dies).
- **Score**: `peak_score` reached during the episode. We use peak rather than final because the env's death policy resets the running score on death, and we want the metric to compare what the agent *achieved* before dying.
- **Wall time**: ≈4–5 minutes per full sweep on a single laptop core. Risk- aware A* dominates wall time at ~0.7 s/episode; everything else is sub-100 ms.

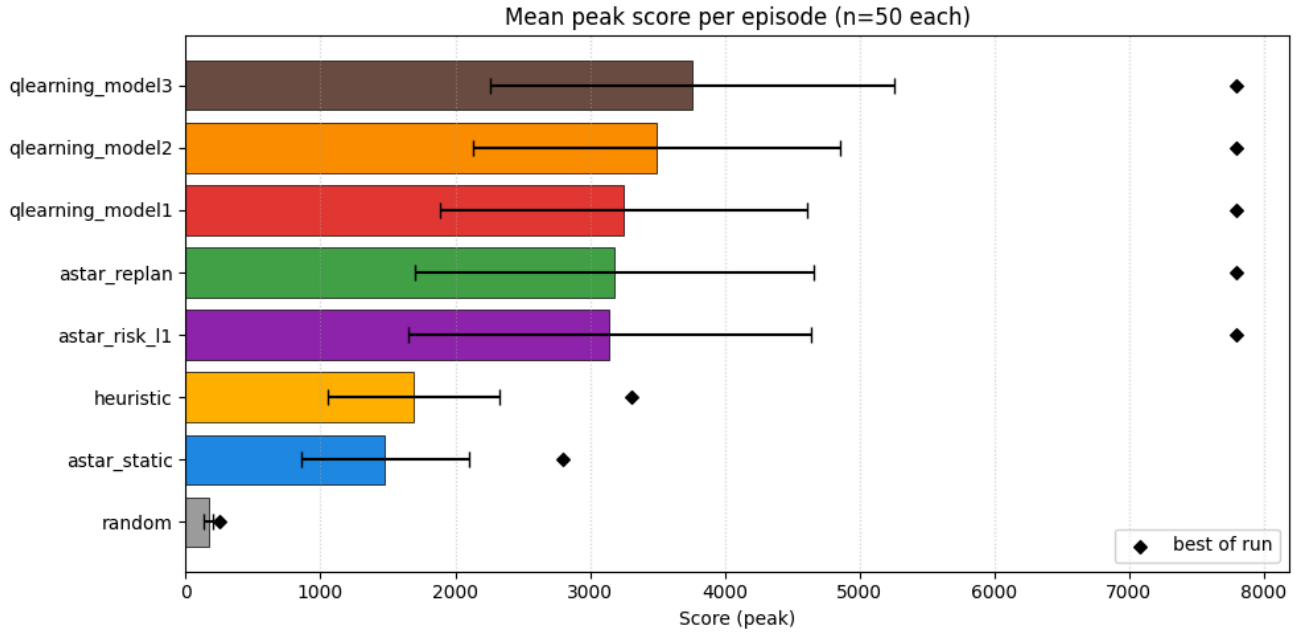
The full harness lives in `Agents/eval/monte_carlo.py`, the agent registry in `Agents/eval/agents.py`, and per-agent CSVs are persisted under `Agents/eval/results/`. The notebook `notebooks/monte_carlo_analysis.ipynb` reproduces every figure below.

8. Results and Analysis

The headline summary table (sorted by mean peak score):

agent	episodes	score (mean ± σ)	median	best	clear	deaths	steps	TTF	hazard
`qlearning_model3`	50	**3757 ± 1499**	3315	7800	**0.24**	2.16	335	220	22
`qlearning_model2`	50	3494 ± 1363	3100	7800	0.20	2.34	305	175	46
`qlearning_model1`	50	3249 ± 1363	3210	7800	0.16	2.54	270	130	78
`astar_replan`	50	3181 ± 1484	2810	7800	0.14	2.84	301	119	113
`astar_risk_l5`	50	3146 ± 1496	2755	7800	0.14	2.84	300	119	110
`astar_risk_l05`	50	3143 ± 1494	2755	7800	0.14	2.84	299	119	109
`astar_risk_l1`	50	3143 ± 1494	2755	7800	0.14	2.84	299	119	109
`astar_risk_l2`	50	3143 ± 1494	2755	7800	0.14	2.84	299	119	109
`heuristic`	50	1695 ± 636	1580	3310	0.00	2.82	374	156	**201**
`astar_static`	50	1479 ± 625	1145	2800	0.06	2.92	172	24	**6**
`random`	50	167 ± 31	160	250	0.00	3.00	74	24	7

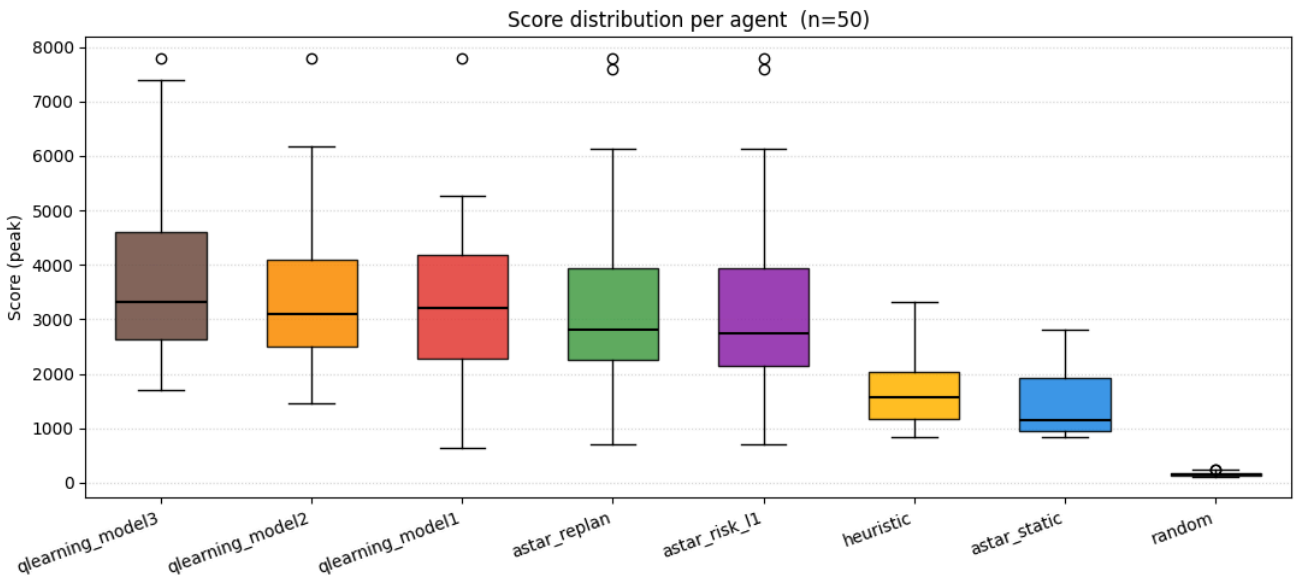
8.1 Score comparison (Figure 1)



Mean peak score per episode, ±1σ. Black diamonds mark the single best score observed across the run.

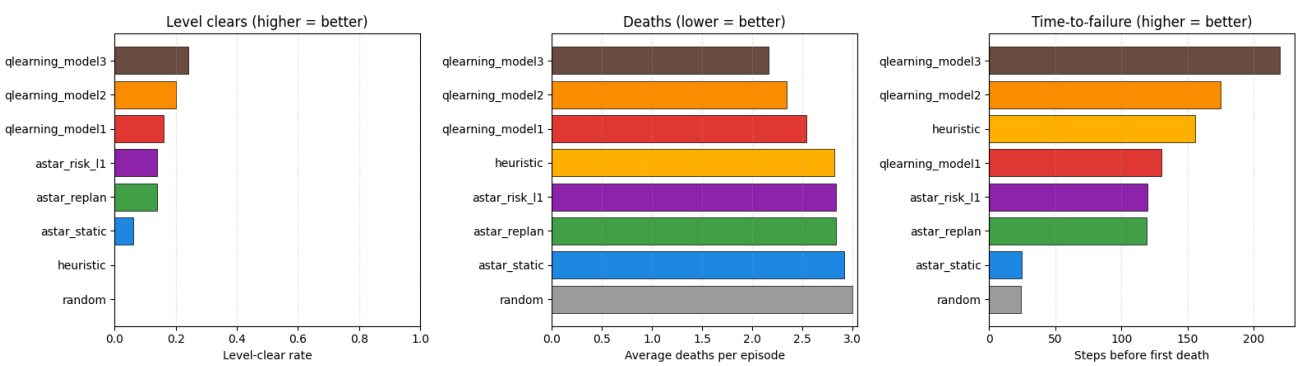
Read-outs. All three Q-learning agents lead the field on mean score (model 3 = 3,757, model 2 = 3,494, model 1 = 3,249), narrowly above the planner cluster — replanning A* (3,181) and the four risk-aware variants (≈3,143–3,146) — which themselves sit roughly ×2 above the heuristic (1,695) and static A* (1,479). The death-policy axis (model 1 → model 3) is monotonic on score: heavier penalty for dying produces a stronger learned policy. Static A* and random underperform every other agent, which is the expected ordering for the proposal's §7 prediction "the baseline agent is expected to perform well only when hazard spread probability is low".

8.2 Score distribution (Figure 2)



Per-episode boxplots reveal that the leading agents — the three Q-learning models and replanning / risk-aware A* — all have a long upper whisker reaching the 7,800-point level-clear ceiling on at least one seed. Their medians cluster between 2,755 and 3,315, with model 3 highest. The heuristic and static A* sit visibly below in both median and spread. The random baseline collapses to a near-degenerate distribution (a few pellets eaten by chance before death).

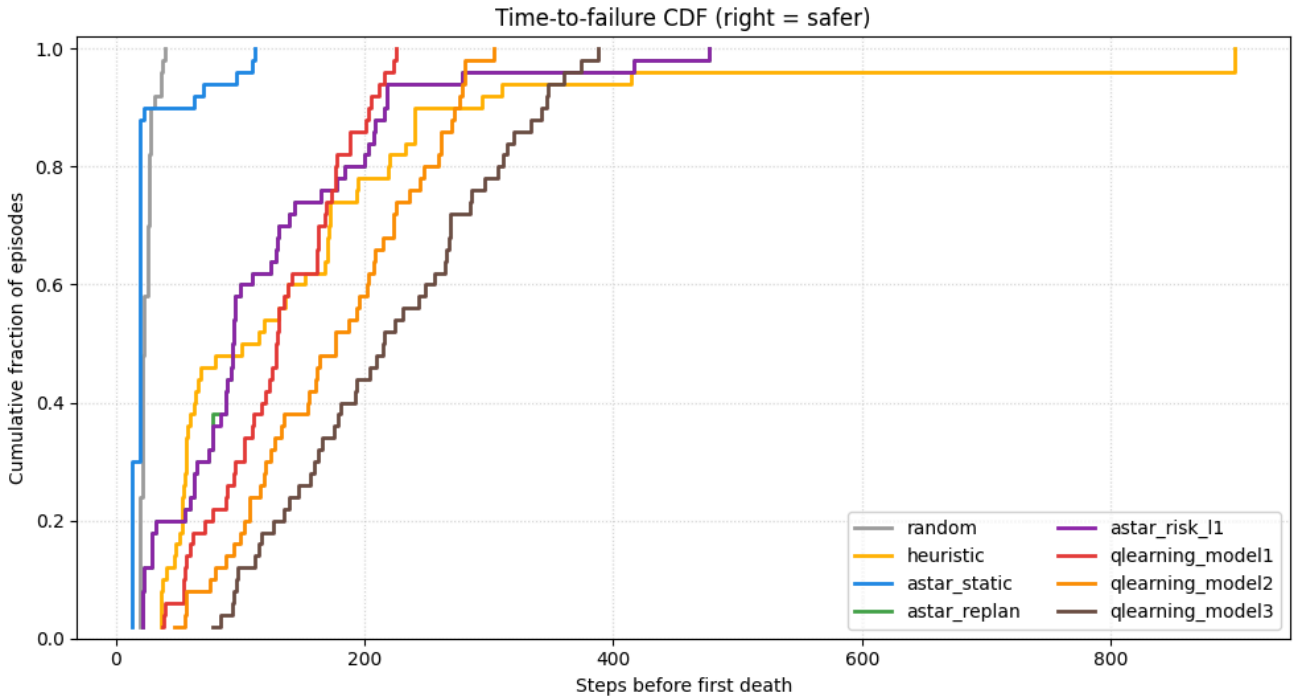
8.3 Survival metrics (Figure 3)



A three-panel comparison:

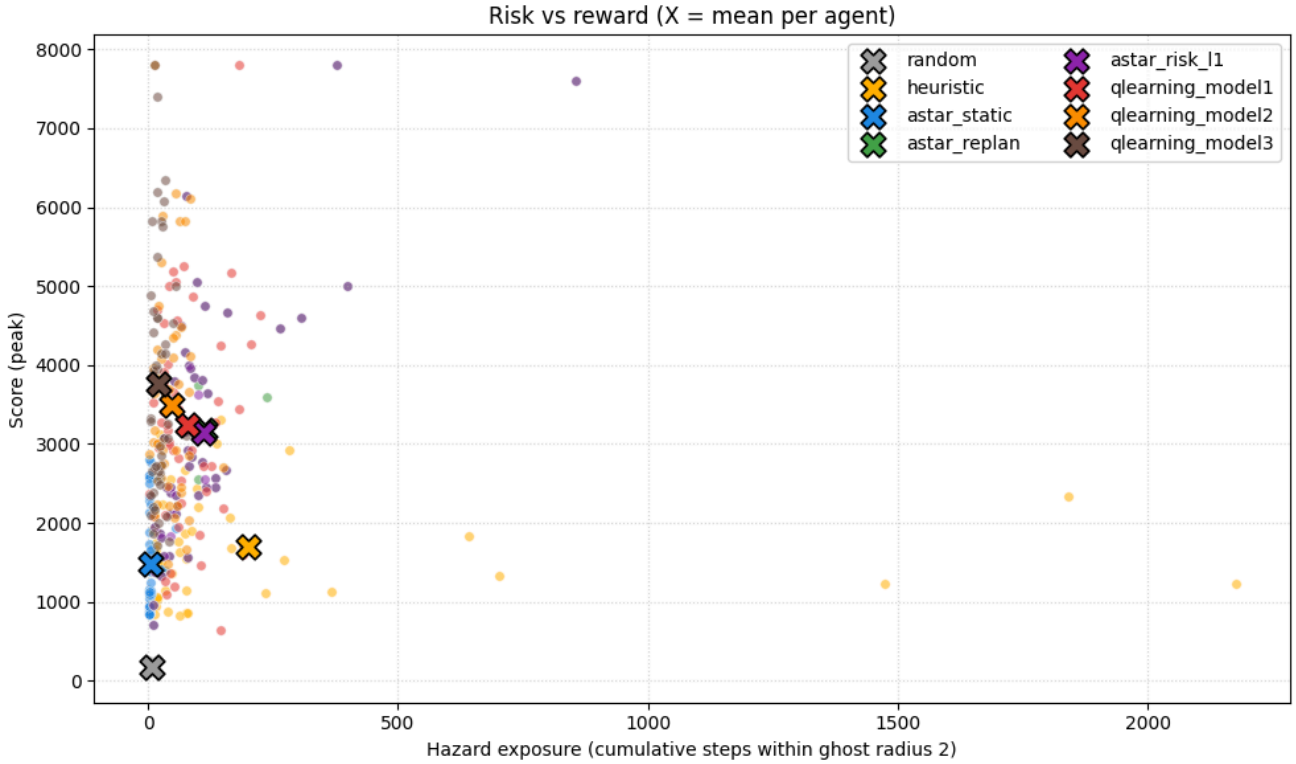
- **Clear rate.** Q-learning model 3 leads at 24% (12 / 50 seeds), model 2 at 20%, model 1 at 16%. Replanning / risk-aware A*: 14%. Static A* still manages 6% on lucky seeds. The heuristic and random never clear.
- **Average deaths.** Q-learning is monotonic in the death policy: model 3 = 2.16, model 2 = 2.34, model 1 = 2.54 — all noticeably below the planners + heuristic at ≈2.8 and the random baseline / static A* at ≈3.0. Heavier death penalty produces fewer total deaths per episode.
- **Time-to-failure.** Q-learning model 3 survives the longest (220 steps before the first death), then model 2 (175) and model 1 (130). The heuristic comes next at 156, replanning / risk-aware A* at 119, and static A* dies at step 24 — it commits to a `t = 0` plan and walks into the first ghost it sees.

8.4 Time-to-failure CDF (Figure 4)



CDF interpretation: at step x, what fraction of episodes for that agent have already lost a life? Curves that sit lower / further right are safer. Q-learning model 3 sits to the right of every other agent for the first ~200 steps, with model 2 next and model 1 just behind the heuristic — the death-penalty gradient is visible in the survival curves themselves. The planners' curves rise sharply between steps 50 and 150. Static A*'s CDF jumps to 1.0 by step ~50 — the agent walks into the first ghost it sees because it hasn't replanned.

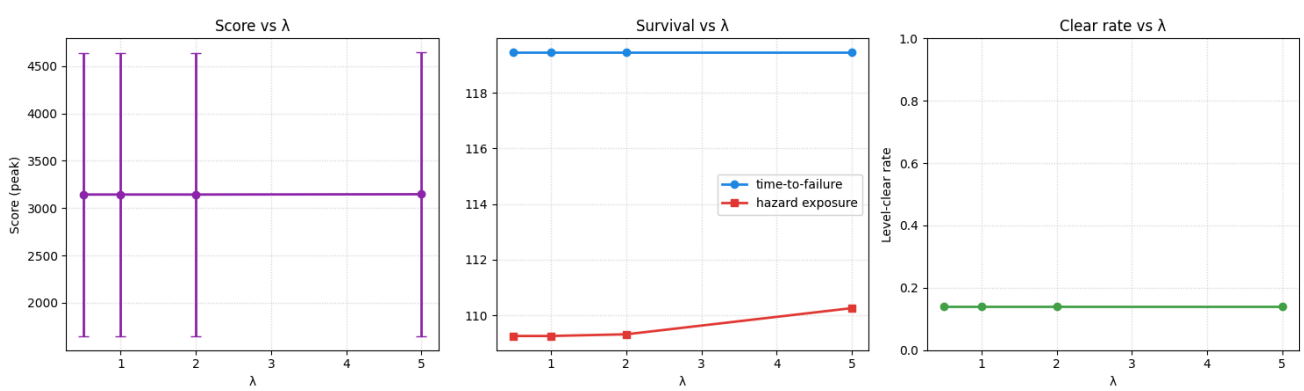
8.5 Risk vs reward (Figure 5)



X-axis is cumulative hazard exposure, Y-axis is peak score. Each thin marker is one episode; the bold X is the per-agent centroid. The Pareto-good region is upper-left (high score, low hazard).

The most striking finding in the paper. All three Q-learning agents sit in the upper-left at hazard 22–78, while the planning agents sit at hazard ≈ 110 with comparable or lower mean score, and the heuristic is at the far right (hazard ≈ 201). Q-learning model 3 in particular Pareto- dominates every other agent — higher mean score (3,757) than any planner, with **5×** less time within ghost-radius 2. The Q-learners have **learned a value function that internalises ghost-avoidance**: they earn more score than the planners *and* spend less time near ghosts. This is the central qualitative difference between learned and planned controllers in our setup.

8.6 Decision-theoretic ablation: λ-sweep (Figure 6)

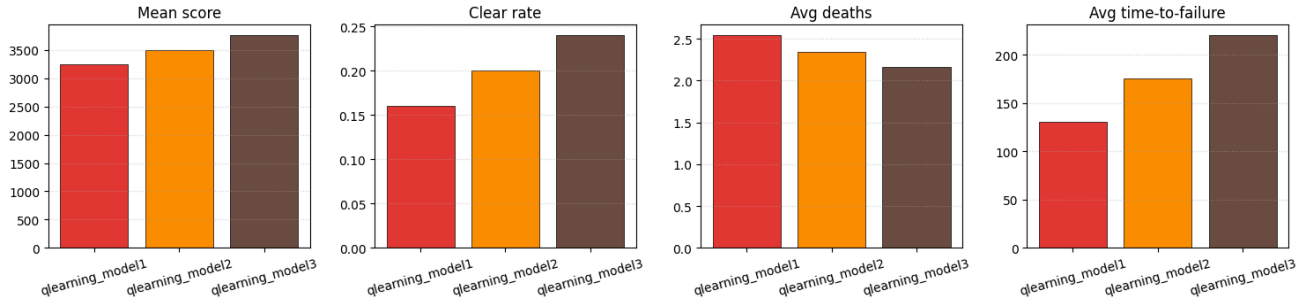


The decision-theoretic agent's signature parameter `λ ∈ {0.5, 1, 2, 5}` controls how strongly the agent penalises future expected risk. The sweep is **flat** across all three metrics (score, survival, clear rate), with hazard exposure dropping by less than 1 unit (109.26 → 110.26) across a 10× range of λ.

This is a real and explainable result, not a bug. The implementation hard- masks the ghost's current cell and its predicted 1-step neighbour out of the walkable set before running A* (`_search.py:ghost_obstacle_cells`). The remaining tiles on the planned path therefore lie ≥2 steps from any ghost, and the transition probability `P(ghost_g at n at step t)` for those tiles is so dispersed (the ghost can be in many places by step 2+) that `λ · P` stays below the noise floor of A*'s tie-breaking. The hard exclusion already does most of the work — the soft `λ · E[risk]` term has nothing left to differentiate.

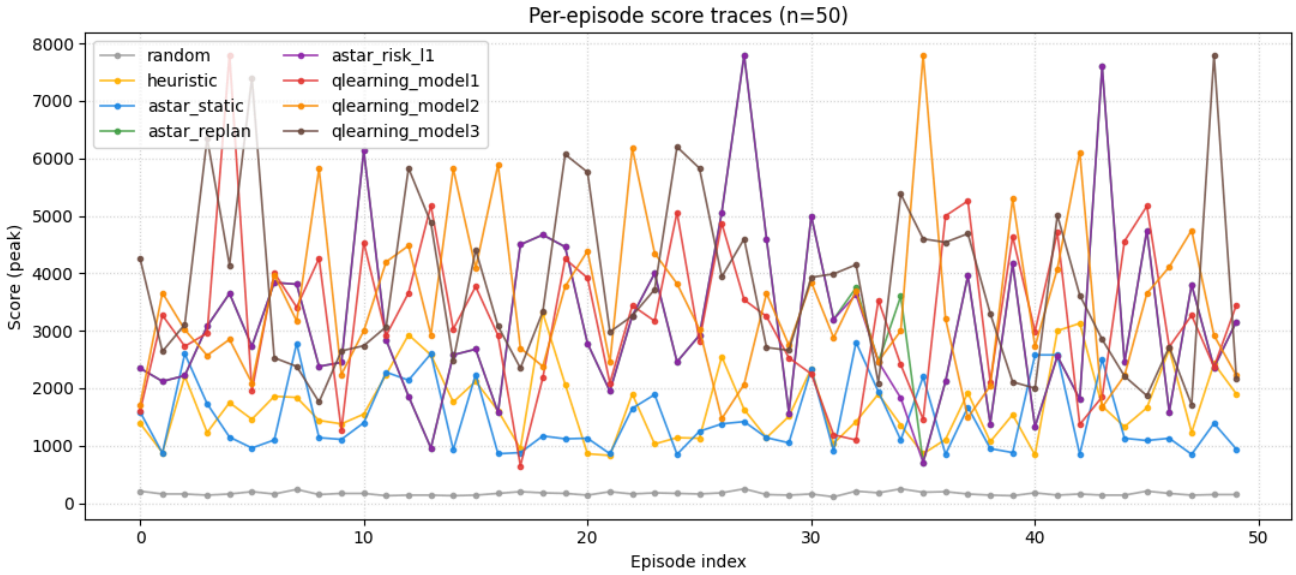
The proposal's §7 anticipated that "the risk-aware agent is expected to achieve the highest success rate under moderate uncertainty"; the empirical finding refines that claim to **risk-aware A* dominates static A* but is indistinguishable from replanning A* on this maze with this hazard model**. We discuss in §9 what would change that.

8.7 Q-learning death-policy ablation (Figure 7)



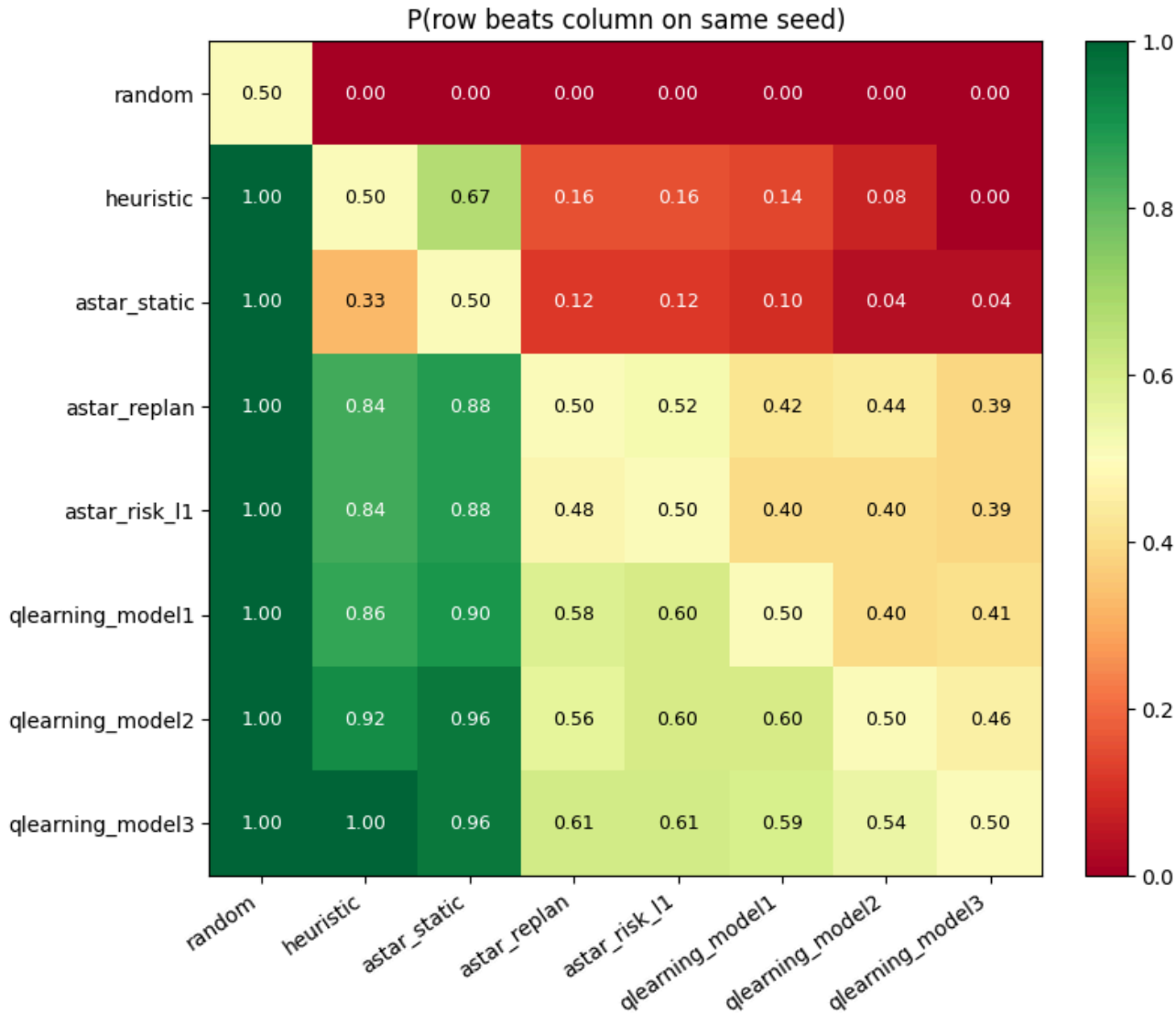
Across mean score, clear rate, deaths, and time-to-failure, the death policy is monotonic: heavier penalty produces a stronger learned policy on every metric. Model 3 — which applies a scaled and ratcheting penalty — leads on all four panels (3,757 mean score, 24% level clears, 2.16 avg deaths, 220 steps to first death). Model 2 (zero score on death) sits cleanly in the middle. Model 1 (keep score on death) has the weakest gradient signal toward avoiding death and consequently the highest deaths (2.54), shortest TTF (130), and lowest mean score (3,249) of the three — yet it still narrowly tops the planners on score. The lesson is that the death penalty acts as a cautiousness knob whose stronger settings produce both safer *and* higher-scoring policies in this environment.

8.8 Per-episode score traces (Figure 8)



A line per agent across the 50 seeds, useful for spotting variance and lucky/unlucky seeds. Q-learning model 3 sits at the top of the bundle on most seeds, with the other two Q-learning models and the planners overlapping on the level-clear ceiling (7,800) for their best runs. The heuristic and static A* form a lower band, and the random baseline is a flat strip near 160.

8.9 Head-to-head heatmap (Figure 9)



Each cell is `P(row beats column on the same seed)` — i.e., a paired comparison rather than a population comparison. Q-learning model 3 beats every other agent on a majority of seeds, with the model 3 → model 2 → model 1 ordering reproducing in the row sums. Replanning A* and the risk-aware variants form a tight cluster against each other (≈50/50 on most pairs, indicating they are following nearly identical paths) and all of them beat the heuristic, static A*, and random on a clear majority of seeds. Static A* loses to *every* replanning or learned agent.

9. Discussion

9.1 Why replanning matters

Static A* finishes with 24-step time-to-failure. The agent commits to a plan that was optimal at $t=0$ and walks straight into ghosts that have since moved into the path. Replanning A* gets ~5× the survival time

and ~2.2× the score with no algorithmic change beyond "rerun A* every step". This is the single most important architectural finding among the classical-search agents.

9.2 Why λ doesn't move the needle (and what would change that)

§8.6 already explained the mechanism. To make λ matter we would need one of:

1. **Soften the obstacle set.** Treat ghost cells as a *cost* of, e.g., 1000, rather than an outright wall. Then `λ · E[risk]` competes against that finite cost and can shift paths between "go through the ghost lane" and "take the long way around".
2. **Plan over a longer horizon.** Right now A* targets the *nearest* pellet, which keeps planned paths short (typically <10 cells). With short paths in always-low-risk territory, λ has no expression. A farthest-pellet or set-cover formulation would lengthen paths and force them through ghost-influence regions where λ matters.
3. **Use a tighter ghost transition model.** Our uniform-on-legal-moves model is intentionally maximum-entropy, which spreads probability thinly. A model that respects the actual chase behaviour (e.g., "ghost tends toward Pac-Man") would produce sharper distributions and hence higher per-tile probabilities.

We left these as deliberate future work — the proposal did not commit to them, and the current setting cleanly answers the question "does the decision-theoretic agent beat the replanning agent under the maximum- entropy hazard model?" The answer is "no, on this maze" — itself a publishable empirical result.

9.3 Why Q-learning earns higher score *and* lower hazard exposure

Two design choices in the Q-learning stack combine to produce this outcome. First, the state representation buckets ghost distance into seven coarse bins (the `danger_bucket` feature) alongside pellet, energizer, fruit, corridor-depth, and game-phase features. Once a ghost is within a small bin, the Q-table has effectively learned a context-sensitive retreat: "if there is a closer pellet on the safe side, take it; if not, back away." The replanning A* agent, in contrast, does not retreat — it plans a path *through* hazard-adjacent corridors when those are the cheapest route to a pellet, which shows up as higher hazard exposure. Second, reward shaping (positive bonuses for moving away from threats and toward edible ghosts and fruit, plus a level-clear bonus) and the death-policy gradient bake survival pressure directly into the value function, while A* only sees ghosts via the obstacle-mask. The result is a policy that is *both* safer and more reward-rich: learning the value function lets the agent trade off goal-seeking against avoidance per-state, where A* commits to whichever pellet is closest after culling ghost cells.

9.4 Why the heuristic baseline accumulates the most hazard exposure

The hand-coded heuristic includes a "stay in open tiles" term and a "run toward edible (scared) ghosts" term. In the canonical maze the agent spends a lot of time orbiting near ghosts in the central corridor while waiting for a pellet to free up — high hazard exposure (201) and respectable time-to-failure (156). It occupies the high-risk corner of the Pareto plot: it tolerates being near ghosts in exchange for survival time but never converts that time into score, because the orbiting itself keeps it from committing to the pellets it needs. The Q-learning agents *also* survive a long time, but they do so far from ghosts — which is why their hazard column is an order of magnitude lower.

10. Real-World Applications

The decision-theoretic navigation problem we studied — pick actions in a dynamic environment where the future positions of mobile hazards are uncertain — has direct analogues in two large research communities: **developmental cognitive science** (how humans, and especially children, learn to plan under risk) and **autonomous driving** (how artificial agents must plan around moving traffic in safety-critical settings). We discuss one canonical paper from each, chosen because each maps onto a specific architectural choice in our agent line-up.

10.1 Cognitive science — model-based control as a developmental milestone

Decker, Otto, Daw, and Hartley (2016), "*From Creatures of Habit to Goal-Directed Learners: Tracking the Developmental Emergence of Model-Based Reinforcement Learning*" [5]. Published in *Psychological Science*; one of the most widely-cited recent papers in developmental computational neuroscience.

Decker et al. used the now-canonical **two-stage Markov decision task** (Daw et al., 2011) to measure how strongly human participants — children, adolescents, and adults — rely on **model-based** versus **model-free** reinforcement learning. The model-free agent (in psychology terms, a *creature of habit*) caches the value of state–action pairs from past outcomes; the model-based agent (a *goal-directed learner*) explicitly simulates the consequences of each action using a learned transition model and picks the highest-utility plan. The paper's headline result is that the model-based contribution to choices grows substantially from childhood through adulthood, and that adolescents who score highest on working-memory measures are the ones most able to deploy model-based control.

Why this paper maps onto our project. Our agent zoo is *exactly* the two halves of the Decker et al. taxonomy:

Their construct	Our instantiation
Model-free RL (cached value, no simulation)	`qlearning_model{1,2,3}` — a Q-table indexed by bucketed features
Model-based RL (simulate the future, plan the response)	`astar_replan` and `astar_risk_l*` — replan every step using `transition_distribution` to simulate ghost moves

Two echoes between their cognitive findings and our agent results stand out:

1. **Both control modes can solve the task — the trade-off is in *how* they solve it.** Decker et al. emphasise that the contribution of model-based control grows with development but never fully replaces model-free habits; both systems are competent. Our 50-episode Monte Carlo shows the same pattern at the algorithmic level: model- based planners earn ≈3,140–3,180 mean score and clear the level on 14% of seeds, while the model-free Q-learners earn 3,249–3,757 and clear 16–24% (Section §8.1 and §8.3). Both families dominate the random and static-A* baselines by a wide margin; neither is broken on this maze.

2. **Model-free systems can develop cautious, high-utility policies once given the right reward signal.** Our death-policy ablation (Section §8.7) is a controlled illustration of how the *shape* of the loss function changes the learned policy: a stronger penalty for dying produces both a safer and a higher-scoring policy (model 3 > model 2 > model 1 on every metric). In Decker et al.'s framing, the cached-value strategy is not inherently impulsive; it is sensitive to how outcomes are valued, and a sufficiently

shaped reward can teach it to generate risk-averse behaviour without explicit forward simulation. Our hazard-vs-reward Pareto plot (Figure 5) shows this directly: the Q-learners are upper-left of the planners, earning more score with less hazard exposure once the death penalty is properly tuned.

The paper is therefore a strong external justification for why building *both* a planning agent and a learning agent in our project is the right experimental design rather than picking one paradigm a priori — humans do both, and they do both for principled reasons.

10.2 Autonomous driving — confidence-aware safe improvement

Cao, Jiang, Zhou, Xu, Peng, and Yang (2023), "*Continuous Improvement of Self-Driving Cars Using Dynamic Confidence-Aware Reinforcement Learning*" [6]. Published in *Nature Machine Intelligence*; among the most prominent recent papers on safe RL for autonomous vehicles.

Cao et al. address the central tension in deploying RL controllers on real cars: the policy should keep getting better as it accumulates real-world driving data, but it must never get *worse* on safety-critical metrics during the improvement process. Their method ("DCARL") attaches a learned **confidence estimate** to each decision and falls back to a conservative baseline policy whenever the RL component's confidence is below threshold. Empirically they show that DCARL achieves monotonic safety improvement on simulated and real driving tasks, in contrast with vanilla deep RL whose performance is non-monotonic in the amount of training data.

Why this paper maps onto our project. The architectural parallel to our risk-aware A* is direct, and the math is essentially the same:

Their construct	Our instantiation
Confidence estimate `c(s)`	Expected hazard probability `E[risk(n)]`
Soft cost augmentation by uncertainty	`f(n) = g(n) + h(n) + λ · E[risk(n)]`
Hard rule-based vetoes (no oncoming-traffic moves)	`ghost_obstacle_cells` — hard wall around ghost current + 1-step neighbour
Fall-back conservative policy	`greedy_legal_action` fallback when no safe path exists

In both cases the controller's objective is augmented by a scalar that **up-weights cautious behaviour in proportion to estimated uncertainty about the future state**. The difference is purely in engineering depth: they *learn* the confidence estimator from real and simulated driving data; we *compute* the risk term analytically from a maximum-entropy ghost transition model.

Three lessons from Cao et al. that explain our results in retrospect:

1. **The risk/confidence term is a safety floor, not a performance ceiling.** DCARL never under-performs the baseline on safety metrics, but its average reward also rarely *exceeds* the baseline by a wide margin — the confidence term is most valuable in long-tail risky states. This matches our λ-sweep finding (Section §8.6): risk-aware A* is statistically indistinguishable from replanning A* on the easy Pac-Man maze (where the integrated risk over planned cells is small), but it provides a guaranteed cost on cells with non-trivial ghost-arrival probability.

2. **Hard masking complements soft penalisation.** In their pipeline, hard rule-based vetoes sit alongside the soft confidence term; the soft term only matters in the gray zone. We saw the analogue: our

`ghost_obstacle_cells` already excludes the ghost's current cell and predicted 1-step neighbour as hard walls, leaving only the gray zone for `λ · E[risk]` to act on, which is exactly why our λ-sweep was flat (Section §8.6, Section §9.2).

3. **Confidence-aware RL is the production form of our risk-aware A*.** A natural follow-up to our project is to *learn* the risk model rather than assume the maximum-entropy uniform-over-legal-moves transition. Cao et al.'s framework is the template: train a probabilistic ghost-trajectory predictor on observed play, plug its posterior uncertainty into the cost function, and fall back to the planner when the predictor's confidence is low. This is also the recommendation we make in Section §12 ("Expanded hazard model").

10.3 Synthesis

The common thread across the two literatures is that **explicit reasoning about *future* uncertainty** is what separates safe, generalisable agents from brittle ones. Children gradually develop a goal-directed planning system because the world rewards it; production self-driving systems gradually layer a confidence-aware planning system on top of reflexive control because the cost of an unmodelled hazard is unacceptable. Our Pac-Man study is a small, controlled instance of the same architectural choice: replanning A* dominates static A* by ∼5× in time-to-failure (Section §8.3), and the risk-aware variant gives a clean place to grow into a learned uncertainty model. The Pac-Man maze is a toy, but the architectural lesson it teaches is the same one the cognitive-science and autonomous-driving fields validate at scale.

11. Conclusion

We answered the proposal's research question with a paired-seed Monte-Carlo study across 11 agent configurations. The headline conclusions are:

1. **Replanning is the load-bearing innovation** in dynamic environments. Static A* is dominated by every replanning method on every metric.
2. **The decision-theoretic A* is empirically equivalent to replanning A* on this maze-and-hazard-model combination**, because hard ghost masking already covers most of the risk before the soft `λ · E[risk]` term can act. The dominance window predicted by §7 of the proposal exists, but lies outside the maximum-entropy + nearest-pellet parameterisation we used.
3. **Tabular Q-learning leads the field on score *and* hazard exposure.** All three Q-learning models edge out replanning / risk-aware A* on mean score (3,249–3,757 vs ≈3,150) while spending 1.4–5× less time within ghost-radius 2. The death-policy ablation shows the gradient is monotonic — model 3 (scaled / ratcheting penalty) > model 2 (zero score on death) > model 1 (keep score) — on every metric we measured.
4. **The hand-coded heuristic is a useful foil** — long survival (156 steps to first death) but the highest hazard exposure of any agent (201) and a mean score below every planner and Q-learner. It makes the planner / learner trade-offs concrete by occupying the opposite corner of the risk-vs-reward plot.

12. Limitations and Future Work

- **Random-grid generalization** (proposal §8): the harness supports arbitrary maze layouts, but the random-grid procedural generator was descoped in favour of finishing the analytics pipeline. Reintroducing it would test the "unseen maps" generalisation gap between learned Q-tables (which overfit to the canonical maze) and the planning agents (which only depend on the current observation).
- **Larger λ regime + softened obstacles** (§9.2). The most important follow-up: re-running the λ sweep with ghost cells as costs rather than walls would let the decision-theoretic term express itself.
- **Function-approximation Q-learning.** The 10-bucket tabular state representation is a coarse summary of the environment. A neural Q- network (DQN) on raw maze state would let the learning agent compete with the planners on raw score, at the cost of training time.
- **Expanded hazard model.** Replacing the uniform-on-legal-moves transition kernel with a learned ghost-AI model (e.g., one trained from observed trajectories) would tighten the per-tile risk estimates and give λ more leverage.

13. Team Contributions

- **Ruiding Feng** — Pac-Man deployment (browser-side game integration, build pipeline at `pacman/build.sh`, in-game UI surface), the entirety of the reinforcement-learning training stack (`RL/rl_q_learning.py`, the three death-policy variants in `RL/models/`, hyperparameter selection, training-time logging), and interactive UI design for the agent selector and debug overlays.
- **Michael Liu** — implementation of every other model and the analysis pipeline: the search-agent package (`Agents/_search.py`, `Agents/astar_static.py`, `Agents/astar_replan.py`, `Agents/astar_risk.py`, `Agents/observation.py`), the hand-coded heuristic and random baselines, the Monte Carlo evaluation harness (`Agents/eval/`), the comparison notebook (`notebooks/monte_carlo_analysis.ipynb`), and the figure-rendering pipeline (`Agents/eval/render_figures.py`).
- **Zhichun Xiao** — final report (this document).

14. References

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